

Physics 111B: Rutherford Scattering

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October 2025

Abstract

$$m = \frac{\Delta V}{\Delta \nu} = \frac{1.78-0.68}{(9.58-5.19) \times 10^{14}} = \frac{1.10}{4.39 \times 10^{14}} = 2.51 \times 10^{-15} \text{ V}\cdot\text{s}$$

Rutherford scattering will be tested using a collimated ^{241}Am source and thin gold foils inside a vacuum chamber. A silicon PN detector and pulse-height analyzer measure energy spectra and count rates at set scattering angles. From the counts versus angle we test the $\sin^{-4}(\frac{\theta}{2})$ angular dependence predicted for Coulomb scattering and use the angular distribution to infer the nuclear charge Z of gold. Control of collimation, foil thickness, and solid angle, along with long acquisitions and background rejection, are required to achieve good statistics at large angles. Results illustrate the nuclear model of the atom.

1 Introduction

The Rutherford scattering experiment provides one of the most direct demonstrations of the nuclear model of the atom. Before Rutherford's work, the atom was thought to be a diffuse positive sphere with embedded electrons, known as the "plum pudding" model. When α particles were directed at a thin gold foil, most passed through with little deflection, but some were scattered at large angles. This unexpected result could only be explained if almost all of the atom's mass and positive charge were concentrated in a tiny nucleus.

In this experiment, α particles from a ^{241}Am source are collimated and directed toward thin gold foils inside a vacuum chamber. A silicon detector measures the energy and count rate of scattered particles at various angles. By comparing the observed angular distribution to the theoretical Rutherford formula, which predicts that scattering probability varies as $\sin^{-4}(\frac{\theta}{2})$, the nuclear charge of gold can be confirmed. The Rutherford formula assumes that the α particle interacts with a single nucleus at a time, the nucleus is effectively stationary and point like, the Coulomb interaction is unscreened, and the kinetic energy of the α particle is not significantly reduced during scattering. Deviations from the theoretical curve can arise from electron screening, multiple scattering, or nuclear size effects. The confirmation of this scattering was decisive evidence for the nuclear model of the atom and it demonstrated that the atom consisted mostly of empty space with a tiny, dense, positively charged nucleus at its center.

In this experiment we measured the relative numbers of α particles scattered as a function of scattering angle. There is a $\sin^{-4}(\frac{\theta}{2})$ dependence that we will use to calculate the nuclear charge of gold.

Experimentally, the number of counts $N(\theta)$ detected in a given time interval at angle θ is proportional to the differential cross section:

$$N(\theta) \propto \frac{d\sigma}{d\Omega} \quad (2)$$

Rearranging it gives an equation for the nuclear charge of gold:

$$Z = K \sqrt{N(\theta)} \sin^2\left(\frac{\theta}{2}\right) \quad (3)$$

where K is a proportionality constant that incorporates the source strength, foil thickness, detector solid angle, and acquisition time.

2 Theory

The theoretical scattering cross-section for Coulomb interaction between an α particle and a gold nucleus is given by

$$\frac{d\sigma}{d\Omega} = \left(\frac{Z_1 Z_2 e^2}{16\pi\epsilon_0 E} \right)^2 \sin^{-4}\left(\frac{\theta}{2}\right) \quad (1)$$

3 Apparatus and Procedure

3.1 Apparatus

The experimental setup used for Rutherford scattering is shown in Figure 1.

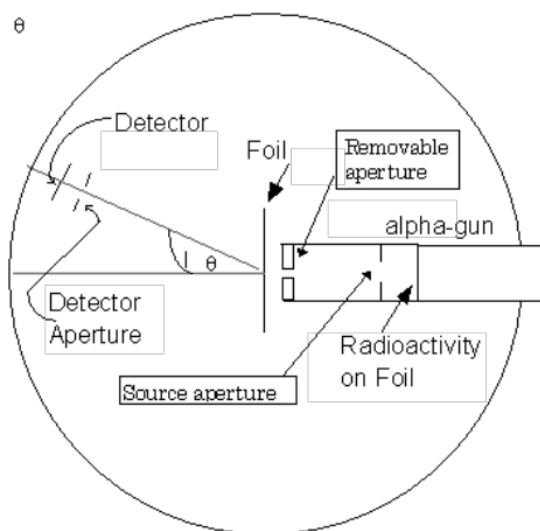


Figure 1: Diagram of Vacuum Chamber showing the gold foil target, detector arm, and collimated alpha source.

Top view of the entire apparatus used in the experiment

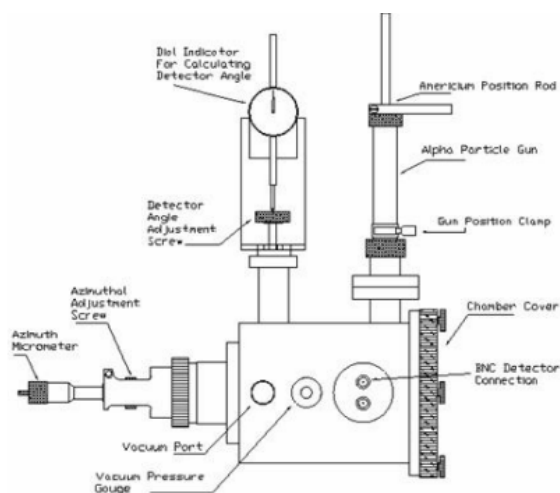


Figure 2: Diagram of Apparatus, Top View.

A collimated ^{241}Am α -source emits 5.486 MeV α particles toward the gold foil. Scattered particles are detected by a silicon PN detector mounted on a rotatable arm, with angle θ measured relative to the beam axis. Count rate data are collected using a pulse-height analyzer for multiple scattering angles.

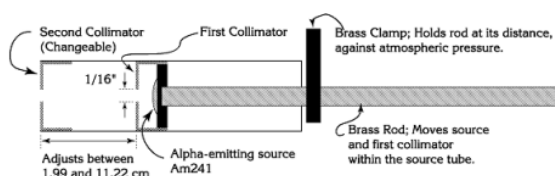


Figure 3: Diagram of the Alpha Gun.

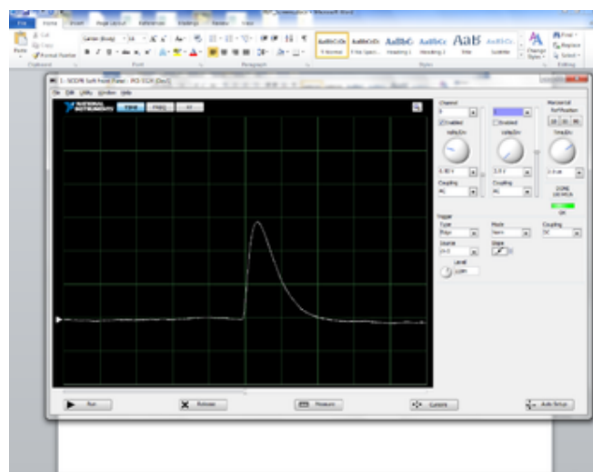


Figure 4: Oscilloscope Signal from PG-2

3.2 Procedure

1. Safety and Pre-checks

- Before starting make sure proper radiation training is done and wear your radiation ring at all times. Begin with surveying the setup with a Geiger counter before beginning and make sure absolutely no food or drink is near the station. Remove the front chamber cover, be careful as the apparatus is very fragile and the detector is light-sensitive.
- Using the detector angle adjusting screw in figure 2, set the detector to 0° on the front protractor.
- Remove the empty foil holder by gently wiggling it side to side and replace the gold foil with a new one provided.
- Review figure 3 and do not remove the collimators and do not loosen the rectangular brass clamp as it prevents the rod from sliding and pushing out collimators.
- To center the gun, loosen the hose clamp and knurled O-ring clamp on the source tube, move the tube to the chamber center, then finger tighten the O-ring clamp then slide the hose clamp flush against it then tighten. This prevents atmospheric pressure from diving the gun into the detector arm when you pump down the apparatus.

2. Pump-down Procedure

- Verify the cover O-ring and sealing surfaces are clean. Replace the cover and hand-tighten the thumb screw evenly.
- Open the vent valve between the chamber and the pump then turn the pump on.
- To vent it or open it later, slowly open the valve first while listening for a hiss, then turn off the pump as a gentle pressure change protects the delicate foil and minimizes oil back-streaming.

3. Electronics

- Using the oscilloscope, we first traced the signal chain and confirmed stable rates such that our graph looked like the one in Figure 4.
- After that is done, make sure the oscilloscope program is closed and open the PHA program window and make sure the sample rate and ground voltage are at adequate values such that you can see the data well. Once the wanted angle is set acquire the data and save it to a folder that is well labeled.
- Repeat this process multiple times for various angles between -10 and 25 degrees with the larger angles being run overnight.

4 Calculations and Data Reduction

Data reduction involves normalizing count rate $N(\theta)$ by the solid angle $\Delta\Omega$ and verifying the angular dependence $N(\theta) \propto \sin^{-4}(\frac{\theta}{2})$.

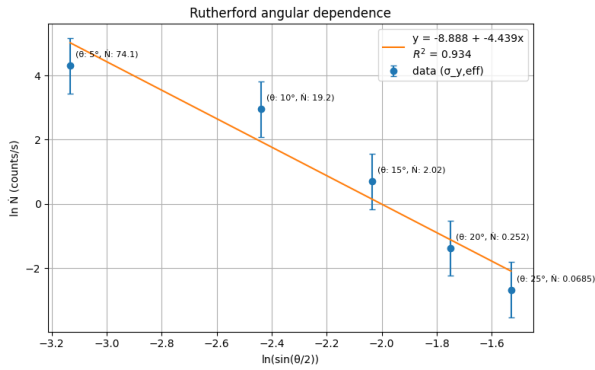


Figure 5: Rutherford Angular Dependence Log-Linearization

We implement the Rutherford law $N(\theta) \propto \sin^{-4} \theta/2$ by turning it into a straight line fit. For each file in an specified angle folder, the code reads the .dat files sums the counts column to get total counts C_{tot} in the combined rate is $N = \frac{C_{tot}}{T_{tot}}$. It then computes $\ln \sin \theta/2$ and $y = \ln(N)$. There are also included error bars.¹ From the files, since only the Live time units are reported we had to convert it using:

$$t_s = \frac{L_t}{0.2427}$$

This is done so that every run is comparable under different exposures.

Since the amount of particles that are read by the sensor is so low at higher angles, we had to do longer tests for the angles the higher they are.

As θ increases from 5 to 25 degrees, the total rate N falls steeply with is what the $N(\theta) \propto \sin^{-4} \theta/2$ predicts. With fewer counts at large θ , the spectra look noisier (Poisson), and the peak amplitude drops by orders of magnitude. There is also

¹For the error bars, their explanation is in the error analysis section

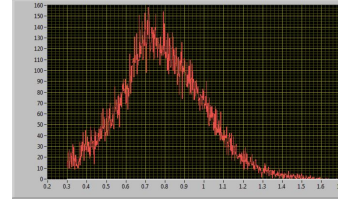


Figure 6: Distribution at 0 degrees

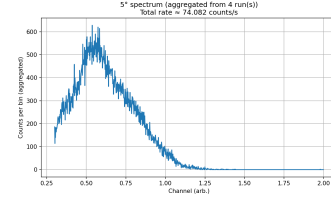


Figure 7: Rutherford Scattering at 5 Degrees

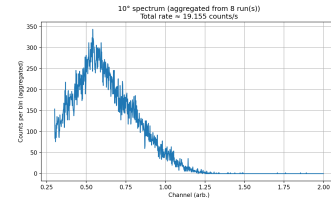


Figure 8: Rutherford Scattering at 10 Degrees

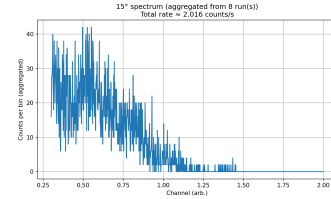


Figure 9: Rutherford Scattering at 15 Degrees

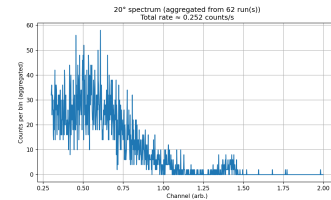


Figure 10: Rutherford Scattering at 20 Degrees

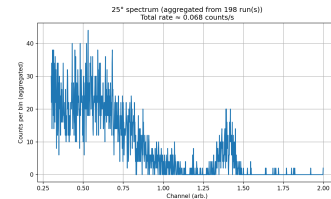


Figure 11: Rutherford Scattering at 25 Degrees

a broadening of the alpha peak towards lower channel with increasing theta as scattered alphas traverse a long path in the foil on exit at larger angles, so energy loss and smaller growth. Any bumps in the 20-25 degree spectra are consistent with low statistics and small backgrounds.

5 Questions

1. For our angle range of 5-25 degrees, screening of the nucleus by electrons is negligible. From the approximation of the Thomas-Fermi angle² we get:

$$a = 0.88a_0 \cdot \frac{1}{Z^{1/3}} \quad (4)$$

where $Z = 79$ for gold, a is $1.1 \cdot 10^{-19}$, and a_0 is the bohr radius. The Born approximation for a screening Yukawa / Thomas-Fermi Coulomb field is:

$$\frac{d\sigma}{d\Omega} = \frac{q^2}{q^2 + \kappa^2} \quad (5)$$

and

$$q = 2k \sin \frac{\theta}{2}, k = \frac{p}{\hbar}, \kappa = \frac{1}{a} \quad (6)$$

This reduces to Rutherford for $\kappa \rightarrow 0$

$$\theta_s \sim \frac{1}{ka} = \frac{\hbar}{pa} \quad (7)$$

For this lab, we use $E = 3.77 \text{ MeV}$ so then we get $p \approx 8.95 \cdot 10^{-19}$ and then finally we get

$$\theta_s \approx 1.079 \cdot 10^{-4} \text{ rad} \approx 0.006177^\circ$$

2. From equation 1, subbing in $Z_1 = 2, Z_2 = 79, E = 3.77 \text{ MeV}$ we can get a k value of $2.28 \cdot 10^2 \text{ MeV fm}$. Subbing that value into the equation:

$$r_0 = \frac{k}{E} \quad (8)$$

we have an R value of $\approx 1.2A^{1/3} \text{ fm} \approx 7 \text{ fm}$ and then we get

$$r_0(3.77 \text{ MeV}) \approx \frac{2.28 \cdot 10^2}{3.77} \\ 60.4 \text{ fm} \gg R$$

Since head-on r_0 is the minimum approach distance, all nonzero-impact-parameter trajectories have:

$$r_{\min} \geq r_0 \gg R$$

Outside the nucleus a spherically symmetric charge produces the same Coulomb field as a point charge, so finite-size effects are negligible across the 5° to 25° range/

3. We don't expect α particles to expect double scattering because of the sheer low possibility of it occurring. For gold, the nuclear radius is about 7 fermis and it has an atomic radius of about 0.13 nm. Therefore, the geometric cross section is about 154 fm^2 , the cross-sectional area of the atom is then about $3.45 \cdot 10^8$ times the area of the nucleus. From this logic, if you were to fire a neutron into a gold atom the probability would be about one in 345 million. For given a scattered particle, the remaining targets would have to cover 1 percent of the area to have a 1 percent probability for a second scattering event, if we use the area ratio, to project a $3.45 \cdot 10^6$ layer of atoms to cover a 1 percent area of the nuclei would result to about 900 micrometers in thickness which is far larger than what we are using for this experiment.

6 Error Analysis

In the log-linear fits we had statistical uncertainty of $\sigma_{y,stat} = \frac{1}{\sqrt{C}}$ then add a single angle-independent systematic f in quadrature to get $\sigma_{y,eff} = \sqrt{\sigma_{y,stat}^2 + f^2}$. Using $\sigma_{y,eff}$ to weight the regression lets the residuals test to model via reduced χ^2 and separates random counting noise from the geometry, gain, alignment systematics. Other sources of error include angle and solid-angle geometry, background subtraction at large θ , human error in reading the distance and the amount of tests we were able to complete for each angle.

7 Raw Data

Since the data points are en masses, the raw data can be seen from the spectrum graphs in the lab report.

8 Conclusion

The Rutherford scattering experiment validates the existence of a dense atomic nucleus and quantitatively confirms the $\sin^{-4}(\frac{\theta}{2})$ dependence predicted by classical Coulomb scattering. Quantitatively, the experimental results demonstrate that the probability of large-angle scattering events is extremely small, consistent with a nuclear diameter several orders of magnitude smaller than the overall atomic radius. This confirms that most of the atomic volume is empty space, with nearly all the mass and positive charge localized in the nucleus. Furthermore, by comparing the measured scattering intensities at various angles to the theoretical Rutherford formula, we can extract an estimate for the effective nuclear charge Z_2 of gold, which closely agrees with its known atomic number of 79.

Deviations from perfect Rutherford behavior at small or large angles can be attributed to experimental and physical effects not accounted for in the idealized model. At small angles, electron screening slightly reduces the effective nuclear charge felt by the α -particles, while at very large angles or higher beam energies, the finite size of the nucleus and multiple scattering

²From the Physics 221B Spring 2021 notes on the Thomas-Fermi Model

events within the foil introduce measurable discrepancies. The experiment not only just reproduces the angular dependence predicted by Rutherford, but also shows the nuclear model of the atom.